COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	09/02/26	**DATE SUBMITTED:**	09/02/26
NAME:	Hare, Andrei Angelo D.	**INSTRUCTOR:**	Engr. Jimlord Quejado

SEATWORK NO. 7.2 Algorithm Analysis: Best-Case and Worst-Case Time Complexity

1. PROCEDURE OUTPUT

What procedures did you perform?

Instructions: Analyze each algorithm carefully. For every item, determine the following:

Best-Case Time Complexity

Worst-Case Time Complexity

Brief explanation of how you arrived at your answers.

Use asymptotic notation such as O(1), O(log n), O(n), O(n log n), O(n^2), etc.

1. Best-Case Time Complexity = O(1) Worst-Case Time Complexity = O(n)
   Explanation: Worst case because it may need to check the elements before it executes while best case because it can check the error immediately

2. Best-Case Time Complexity = O(n) Worst-Case Time Complexity = O(n)
   Explanation: It needs to check all elements, so both cases are linear.

3. Best-Case Time Complexity = O(n) Worst-Case Time Complexity = O(n^2)
   Explanation: In the best case it can only pass through the elements once

4. Best-Case Time Complexity = O(n) Worst-Case Time Complexity = O(n^2)
   Explanation: The algorithm checks elements and repeated operations can increase the time at the worst case scenario

5. Best-Case Time Complexity = O(n) Worst-Case Time Complexity = O(n^2)
   Explanation: The inner loop can repeat for every element of the outer loop

6. Best-Case Time Complexity = O(n) Worst-Case Time Complexity = O(n^2)
   Explanation: The loops process the elements and more repetitions can happens in the worst case scenario

7. Best-Case Time Complexity = O(1) Worst-Case Time Complexity = O(log n)
   Explanation: The algorithm can finish immediately, but dividing the input by 2 repeatedly gives log n

8. Best-Case Time Complexity = O(n) Worst-Case Time Complexity = O(n²)
   Explanation: the loops can cause more operations when the input becomes bigger

9. Best-Case Time Complexity = O(1) Worst-Case Time Complexity = O(1)
   Explanation: It can find the needed value in the first check, but may check still the elements

10. Best-Case Time Complexity = O(n) Worst-Case Time Complexity = O(n²)
    Explanation: The search can finish earlier but the

11. Best-Case Time Complexity = O(n log n) Worst-Case Time Complexity = O(n log n)
    Explanation: The algorithm combines both linear and logarithmic operations

12. Best-Case Time Complexity Worst-Case Time Complexity
    Explanation:

13. Best-Case Time Complexity = O(1) Worst-Case Time Complexity = O(1)
    Explanation: It directly gets the maximum value without checking many elements

14. Best-Case Time Complexity = O(n3) Worst-Case Time Complexity = O(n3)
    Explanation: The three nested loops can each run n times

15. Best-Case Time Complexity = O(1) Worst-Case Time Complexity = O(log n)
    Explanation: It can find the target immediately while repeated reduction makes it logarithmic

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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